Ambiguity-Aware Visual-Language Modeling for Single Answer Grounding in VQA

Jonathan Xu
University of Maryland
College Park, MD
jonxu@terpmail.umd.edu

Matt Hsieh
University of Maryland
College Park, MD
mhsieh05@terpmail.umd.edu

Abstract

In general, visual question answering is considered to be a single answer problem that takes an image question pair as input and outputs its single ground truth answer. In reality, however, more than one person can provide a valid response to a visual question because they focus on various parts of the image. This project considers the VQA AnswerTherapy Single Answer Grounding problem, where the challenge is to detect whether all the valid responses to a visual question have the same visual grounding. We implement a practical ambiguity aware classifier using a frozen CLIP vision language encoder and a lightweight multilayer perceptron. The model combines CLIP image features, CLIP question features, image question interaction features, cosine similarity, and grid based image salience statistics designed to capture whether visual evidence is concentrated or dispersed. On the validation set, the model reaches an F1 score of 0.9361 after threshold tuning, compared with 0.9079 for the majority baseline. Although the model does not fully implement the original proposal's OCR and object proposal branches, the results show that pretrained vision language features plus simple ambiguity sensitive image features can improve single grounding prediction. We also include a diagnostic analysis of ground truth polygon dispersion, showing that multiple grounding examples have much larger average centroid distances than single grounding examples.

1 Introduction

The typical interpretation of VQA is a scenario where the task is to predict one answer to the question posed on the basis of an image (4). This setup is useful, but it hides an important issue: different annotators can provide different valid answers to the same question since they have grounded their answers based on different aspects of the image. For example, if an image contains several similar products and the question asks "What does this say?", one annotator may ground the answer in one label while another may ground a different answer in another label. In that case, disagreement is not just random noise. It reflects real ambiguity in how the image and question connect.

This project focuses on the VQA AnswerTherapy Single Answer Grounding task (1). Given an image question pair, the task is to predict whether all valid answers share a single grounding region or whether valid answers correspond to multiple different grounding regions. The model outputs a score between 0 and 1, where higher values indicate that the example is predicted to have a single common grounding.

This problem matters because ambiguity detection can make VQA systems more trustworthy. A model that knows when a question has multiple possible visual groundings can avoid presenting one answer as if it were certain. This is especially important for accessibility centered datasets such as VizWiz, where images are collected from blind or visually impaired users and may be blurry, cluttered, poorly framed, or text heavy (3; 2).

Preprint.

Our contribution is a practical ambiguity aware visual language baseline. Instead of training a large model from scratch, we freeze a pretrained CLIP encoder (6) and train a small MLP classifier on top of extracted features. The feature vector includes: (1) CLIP image embeddings, (2) CLIP text embeddings for the question, (3) elementwise image text products, (4) absolute image text differences, (5) cosine similarity, and (6) lightweight grid region salience statistics. The grid statistics are intended to approximate the original region competition idea without requiring a full object detector or OCR pipeline.

The main findings are:

- Majority baseline is a strong starting point because the validation dataset has an imbalance towards single grounding examples.
- The frozen CLIP + MLP model improves validation F1 from 0.9079 to 0.9361 after threshold tuning.
- At the default threshold of 0.5, the model has lower F1 than the majority baseline because it sacrifices too much recall, showing that threshold tuning is important for this task.
- A polygon diagnostic shows that multiple grounding examples have much larger average grounding centroid distance than single grounding examples, supporting the hypothesis that spatial dispersion is related to answer ambiguity.

2 Related Work

The closest related work is VQA Therapy (1). This paper introduces a dataset called VQA AnswerTherapy that addresses answer disagreements using visual grounding techniques. The approach redefines answer disagreements from being just label noise to being grounded multimodal discrepancies. Our project uses their Single Answer Grounding task and focuses on predicting whether answer groundings collapse to one region or spread across multiple regions.

This task also builds on earlier work on grounding answers for visual questions asked by visually impaired people (2).The importance of this research lies in the fact that the VizWiz images come from real users, not clean benchmark style images. As a result, the data includes blur, unusual viewpoints, clutter, and text heavy scenes. These properties make ambiguity detection more realistic and more difficult.

The VizWiz benchmark introduced VQA in an accessibility setting, where blind or visually impaired users ask questions about their own images (3). This differs from standard VQA because the images are often captured under uncontrolled conditions and the questions are driven by actual needs for information.

The original VQA dataset and formulation (4) helped establish image question answering as a standard vision language benchmark. However, standard VQA usually evaluates whether a model predicts the expected answer, while our task evaluates whether valid answers are grounded in one shared region.

Our model relies on CLIP, a pretrained vision language model trained to align images and text using contrastive learning (6). CLIP is useful here because the project dataset is relatively small, so training a full multimodal model from scratch would likely overfit. Instead, we use CLIP as a frozen feature extractor and train only a small classifier.

Our original proposal mentioned ViLT as another possible vision language encoder. ViLT removes convolutional and region supervision components and directly applies transformer style multimodal modeling to image patches and text tokens (5). In this implementation, we use CLIP instead because it is straightforward to run as a frozen feature extractor and gives strong image text representations without heavy fine tuning.

3 Method

3.1 Task Formulation

Each example consists of an image, a question, and a binary label. A label of 1 means the example has a single answer grounding, while a label of 0 means valid answers may correspond to multiple grounding regions. The model predicts a score $s \in [0, 1]$, interpreted as confidence that the example has a single shared grounding.

3.2 Frozen CLIP Feature Extraction

We use `openai/clip-vit-large-patch14` as a frozen vision language encoder. For each image question pair, CLIP produces an image embedding v and a text embedding t. Both embeddings are normalized to unit length. From these embeddings, we construct several interaction features:

$$[v, t, v \odot t, |v - t|, \cos(v, t)].$$

The image embedding captures visual content, the text embedding captures the question, the element-wise product and absolute difference represent cross modal interactions, and the cosine similarity provides a direct measure of image question alignment.

3.3 Lightweight Grid Region Salience Features

The original project proposal focused on region competition: single grounding examples should have one dominant visual region, while multiple grounding examples may have several plausible regions. The implemented system approximates this idea using a lightweight 4×4 image grid rather than a learned object detector.

Each image is resized to 224×224 and divided into 16 grid patches. For each patch, the code computes simple visual statistics: grayscale contrast, saturation, and edge strength. These values are summed into a patch salience score. A softmax over the 16 patch scores gives a salience distribution over the image grid.

From this distribution, the model computes:

- normalized entropy of the salience distribution,
- gap between the top two patch salience weights,
- fraction of patches with above average salience,
- spatial variance of the salience distribution,
- mean contrast, saturation, and edge strength,
- image found and image missing indicators.

These features are not a full grounding model, but they give the classifier some information about whether the image evidence appears concentrated or visually spread out.

3.4 Classifier and Training

The final input feature vector is the concatenation of the CLIP based features and the grid region salience features. Before training, features are standardized using the training split statistics and then applied to validation and test features.

The classifier is a multilayer perceptron with two hidden layers of dimension 256 and 128, respectively. It uses LayerNorm after the first linear layer, GELU activations, dropout rates of 0.2 and 0.1, and a final scalar output logit. The model is trained with binary cross entropy with logits. Because the data is imbalanced toward single grounding examples, training uses class balanced weights. We optimize using AdamW with learning rate 0.001 and weight decay 0.0001.

The model is trained for up to 80 epochs with early stopping based on validation F1, using a patience of 12 epochs. Since the official metric emphasizes F1, we tune the decision threshold on the validation set rather than always using 0.5.

4 Experiments

4.1 Dataset

We use the VQA AnswerTherapy annotations associated with the Single Answer Grounding challenge (1; 7). The dataset combines examples from VizWiz and VQA. The loaded training split contains

3,794 examples, the validation split contains 646 examples, and the test split contains 1,385 examples. Labels are available for train and validation, while the test split is used to produce a submission JSON.

Table 1: Dataset split statistics computed from the provided annotation files. The validation set is imbalanced toward single grounding examples, which makes the majority baseline difficult to beat in F1.

Source	Split	Examples	Single	Multiple	Unlabeled
VizWiz	Train	2146	1713	433	0
VizWiz	Val	407	309	98	0
VizWiz	Test	889	0	0	889
VQA	Train	1648	1542	106	0
VQA	Val	239	228	11	0
VQA	Test	496	0	0	496

The full validation set has 537 single grounding examples and 109 multiple grounding examples. This class imbalance is important because predicting the majority class for every example already gives a high F1 score.

4.2 Baseline

The main baseline predicts the majority class for every validation example. Since most validation examples are single grounding, this baseline predicts single grounding for all inputs. It achieves perfect recall for the single class but zero true negatives for multiple grounding examples.

4.3 Metrics

We report precision, recall, F1, and accuracy. The main metric is F1 because it is the primary challenge metric and balances precision and recall. We report results both at the default threshold of 0.5 and at the best validation threshold.

4.4 Results

Table 2: Validation results for the majority baseline and frozen CLIP + MLP model. At the default threshold, the CLIP model has higher precision but lower recall than the majority baseline. After validation threshold tuning, the CLIP model improves F1 from 0.9079 to 0.9361.

Model	Threshold	Precision	Recall	F1	Accuracy	TP	FP	FN	TN
Majority baseline	0.5000	0.8313	1.0000	0.9079	0.8313	537	109	0	0
Frozen CLIP + MLP	0.5000	0.9543	0.7784	0.8574	0.7848	418	20	119	89
Frozen CLIP + MLP	0.2300	0.8947	0.9814	0.9361	0.8885	527	62	10	47

At the default threshold of 0.5, the CLIP model is more precise than the majority baseline but has lower recall, which hurts F1. After threshold tuning, the model reaches the best validation F1 of 0.9361, improving over the majority baseline F1 of 0.9079.

5 Analysis

5.1 Why Threshold Tuning Matters

The default threshold of 0.5 is not ideal for this task. Since the validation set is strongly imbalanced toward single grounding examples, the F1 optimal threshold is much lower: 0.1143. Lowering the threshold increases recall from 0.8566 to 0.9814, while precision remains reasonably high at 0.8947. This improves F1 from 0.8770 to 0.9361.

This shows that the model's ranking of examples is more useful than its raw calibration at 0.5. For the challenge setting, where F1 is the main metric, threshold selection is an important part of the experimental setup.

5.2 Ground Truth Polygon Dispersion Diagnostic

We also ran a diagnostic using the validation ground truth polygons. This diagnostic is not used as an input feature for the model because it would not be available at test time. Instead, it checks whether the task intuition is reasonable.

Table 3: Ground truth polygon dispersion analysis on the validation set. Multiple grounding examples have much larger centroid distances even though the average number of polygons is similar. This supports the idea that ambiguity is related to spatially separated visual evidence, not simply the number of polygons.

Label	Count	Mean Number of Polygons	Mean Centroid Distance
Single	537	2.1285	0.0005
Multiple	109	2.1376	0.1221

The mean number of polygons is almost the same for single and multiple examples, but the mean centroid distance is much larger for multiple-grounding examples. This suggests that ambiguity is not simply about having more polygons. It is more about whether the evidence regions are spatially separated. This supports the central idea of the proposal where multiple answer groundings are associated with dispersed evidence across the image.

5.3 What the Model Appears to Learn

The tuned CLIP + MLP model improves over the majority baseline because it does not simply predict single grounding for every example. It identifies some multiple grounding cases while keeping high recall for single grounding cases. The model's improvement is relatively small because the majority baseline is already strong, but the confusion matrix shows a meaningful difference: the tuned model finds 37 true negatives compared with 0 for the majority baseline.

However, the model still misclassifies 72 multiple grounding examples as single grounding and 17 single grounding examples as multiple grounding. This suggests that CLIP global features and simple grid salience are helpful but not enough to fully solve the ambiguity problem.

6 Discussion

The biggest limitation is that the final implementation is a simplified version of the original proposal. The proposal described a full Ambiguity Aware Region Competition Network with candidate region proposals and an OCR aware branch. The implemented model instead uses frozen CLIP features and lightweight grid region salience features. This was a practical design choice because object proposal integration and OCR matching would add significant engineering complexity.

A second limitation is that the model uses simple handcrafted salience statistics rather than learned question conditioned region attention. The grid features capture visual concentration, contrast, saturation, and edge strength, but they do not know which regions are actually relevant to the question. For example, a visually salient patch may be irrelevant background, while a small low contrast text region may contain the answer.

A third limitation is that the reported results come from a single saved run. Future experiments could run multiple seeds and report the mean and standard deviation of validation F1. This would make the comparison more reliable.

Another limitation is calibration. The model performs poorly at the default 0.5 threshold but improves after validation threshold tuning. This suggests that the raw output probabilities should not be interpreted as perfectly calibrated confidence values.

Future experiments could implement the original OCR aware branch, since text heavy ambiguity is common in VizWiz style images. A stronger version could use OCR tokens, bounding boxes, and question token similarity to detect when multiple text regions could plausibly answer the same question. Another extension would replace the handcrafted grid features with learned region proposals or segmentation masks. Finally, the moonshot direction would be to produce explanation maps showing which regions caused the model to predict ambiguity.

7 Conclusion

This project studied single answer grounding prediction in VQA AnswerTherapy. The task asks whether all valid answers to a visual question share the same grounding region, making it a useful benchmark for ambiguity aware vision language modeling. We implemented a frozen CLIP + MLP classifier with image question interaction features and lightweight grid region salience statistics. On the validation set, the model achieved an F1 score of 0.9361 after threshold tuning, outperforming the majority baseline score of 0.9079. The polygon diagnostic further supported the motivation behind the task: multiple grounding examples have much larger spatial dispersion between grounding centroids than single grounding examples. Overall, the project shows that even a lightweight pretrained vision language approach can improve ambiguity detection, while also leaving clear room for stronger region level and OCR aware modeling.

8 Disclosure

We used LLMs as a coding support tool for occasional debugging, syntax clarification, and implementation guidance. All final code and written report content were completed and reviewed by our team.

References

[1] Chongyan Chen, Samreen Anjum, and Danna Gurari. (2023) VQA therapy: Exploring answer differences by visually grounding answers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV).*

[2] Chongyan Chen, Samreen Anjum, and Danna Gurari. (2022) Grounding answers for visual questions asked by visually impaired people. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).*

[3] Danna Gurari, Qing Li, Abigale Stangl, et al. (2018) VizWiz grand challenge: Answering visual questions from blind people. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).*

[4] Aishwarya Agrawal, Jiasen Lu, Stanislaw Antol, et al. (2015) VQA: Visual question answering. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV).*

[5] Wonjae Kim, Bokyung Son, and Ildoo Kim. (2021) ViLT: Vision-and-language transformer without convolution or region supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML).*

[6] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. (2021) Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML).*

[7] VizWiz. (2026) Grounding all valid answers for VQA: Single answer grounding challenge. `https://vizwiz.org/tasks-and-datasets/vqa-answer-therapy/`.